impression of a large LLM. Two ongoing proof-of-concept projects using this technique—a conversational AI agent for submitting consumer grievances to the Department of Consumer Affairs and a customer complaint agent for the Indian Railways—are funded by Facebook's parent company Meta and built on its open-source Llama model.

THE AGE OF AGENTS

According to Nasscom vice-chairperson Srikanth Velamakanni, who has co-founded the AI firm Fractal Analytics, "there are two games that India can play" in this field. "Number one, we can play the foundational model game, because maybe a better answer for training models has not been found yet," he says. "Or, we can take the native intelligence that's coming out of a foundational model and do amazing post-training that can be used in all kinds of apps in the world."

What Velamakanni is pointing to is the emerging market for AI products. Right now, LLMs can do a pretty good job in summarisation, creative writing and search tasks. But there are limitations to what else they can do, simply because they aren't trained for those tasks. "We can train them to do much more sophisticated tasks like reasoning, planning and generating workflows," says Velamakanni. "The possibilities are endless. We are just scratching the surface."

Mumbai-based entrepreneur Rustom Lawyer agrees. Starting with a medical transcription company two decades ago, he ventured into voice-based AI in healthcare in 2020 with his startup Augnito. One of its first products, for which he collaborated with IIT's Bhattacharyya, was an automatic speech recognition and natural language processing tool to assist radiologists in writing their reports faster—and thereby being able to spend more time on diagnosis.

Lawyer says the firm went a step ahead with its next product. Using ambient listening technology, its AI suite can translate a doctor-patient conversation, summarise and structure it, while also providing financial data that an insurance company may need to extract. It saves doctors the burden of writing reports that take up a lot of their time. "Think of it as an invisible agent who understands and gets the job done," says Lawyer, adding that the tech has been deployed in 300-plus hospitals in India besides healthcare facilities in the Gulf region and the UK. "Healthcare," he says, "is going to be one of the first areas that actually benefit in a huge way from the AI wave."

The likeliest form in which GenAI would reach consumers across sectors—from finance and legal services to customer support and logistics—is agents. Think of them as the next-gen version of digital assistants such as Apple's Siri or Amazon's Alexa, except that they can make decisions, solve problems and act autonomously. "This year, I expect we will see the rise of AI agents," says Velamakanni, adding that startups are already building online marketplaces where agents performing different tasks will be made available. "Earlier, we used to sell apps. Now you can sell an agent."

THE WAY AHEAD

The Union government on its part is giving a Rs 10,300 crore push towards reducing the cost of innovation through its IndiaAI Mission. Being implemented by MeitY, it will be working on all three layers of AI innovation—chip design, foundational models and applications. According to Union minister of electronics and information technology Ashwini Vaishnaw, 18,000 GPUs are being empanelled under a common computing facility to help startups and researchers nationwide. "The global benchmark is $2.5-3 per GPU hour. The price we have come up with is just above $1, and we are also giving a 40 per cent subsidy on it," he said during his visit to Bengaluru on February 15. "So, the cost of innovation should really reduce, and more and more people should get the opportunities for innovation."



"We should have a skills revolution. We may have a large population but the number of people really skilled in the high-end capabilities required for AI, etc. is very small"

RAJESH NAMBIAR
President, Nasscom

At last count, the country had about 230 startups working on GenAI. "India is still one of the largest AI talent pools globally," says Nambiar. "So, to maximise the potential, we should have a skills revolution. For, we may have a large population and talent space, but the number of people who are really skilled in the high-end capabilities required for AI, cybersecurity, quantum computing, etc. is very small." At the same time, there's also a bid to woo Indian-origin AI experts—who form a sizeable chunk of the global talent pool—back home to tap this emerging opportunity. But, for that, the investment into AI needs to be ramped up. "It's not that we are at the 11th hour. This is just the beginning...it's the first chapter of the book. That's why this conversation has become so much more alive and kicking now," notes Velamakanni. There's clearly a long way to go, and DeepSeek's emergence has shown that the race is far from over. ■